



LectureAssist: An Agentic, Multimodal AI Assistant that Turns Lectures into Adaptive Study Material

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Motivation & Problem

Students re-watch hours of recorded lectures to find a single concept, and have no fast way to self-test on what the teacher actually emphasised.

- Lecture knowledge is locked inside **video, speech and slides** — rarely available as clean text.
- Generic quiz tools **hallucinate** off-topic, ungrounded questions.
- One-size-fits-all difficulty ignores **each student's weak spots**.

Goal. An offline-capable system that *reads the board, hears the lecture*, and produces *grounded, validated and adaptive* study tools at **zero API cost**.

Objectives

- Recover every modality (vision, speech, documents) and time-align it.
- Keep all generated content **anchored** to the real lecture via RAG.
- Orchestrate cooperating LLM agents that **plan, generate, validate and refine** each artifact.
- Adapt difficulty to the learner and run entirely on a **free Colab T4 GPU** with a **local** model.

End-to-End Pipeline

A lecture video or document flows left-to-right through extraction, a grounded knowledge base, the agent orchestrator, and finally validated study outputs.

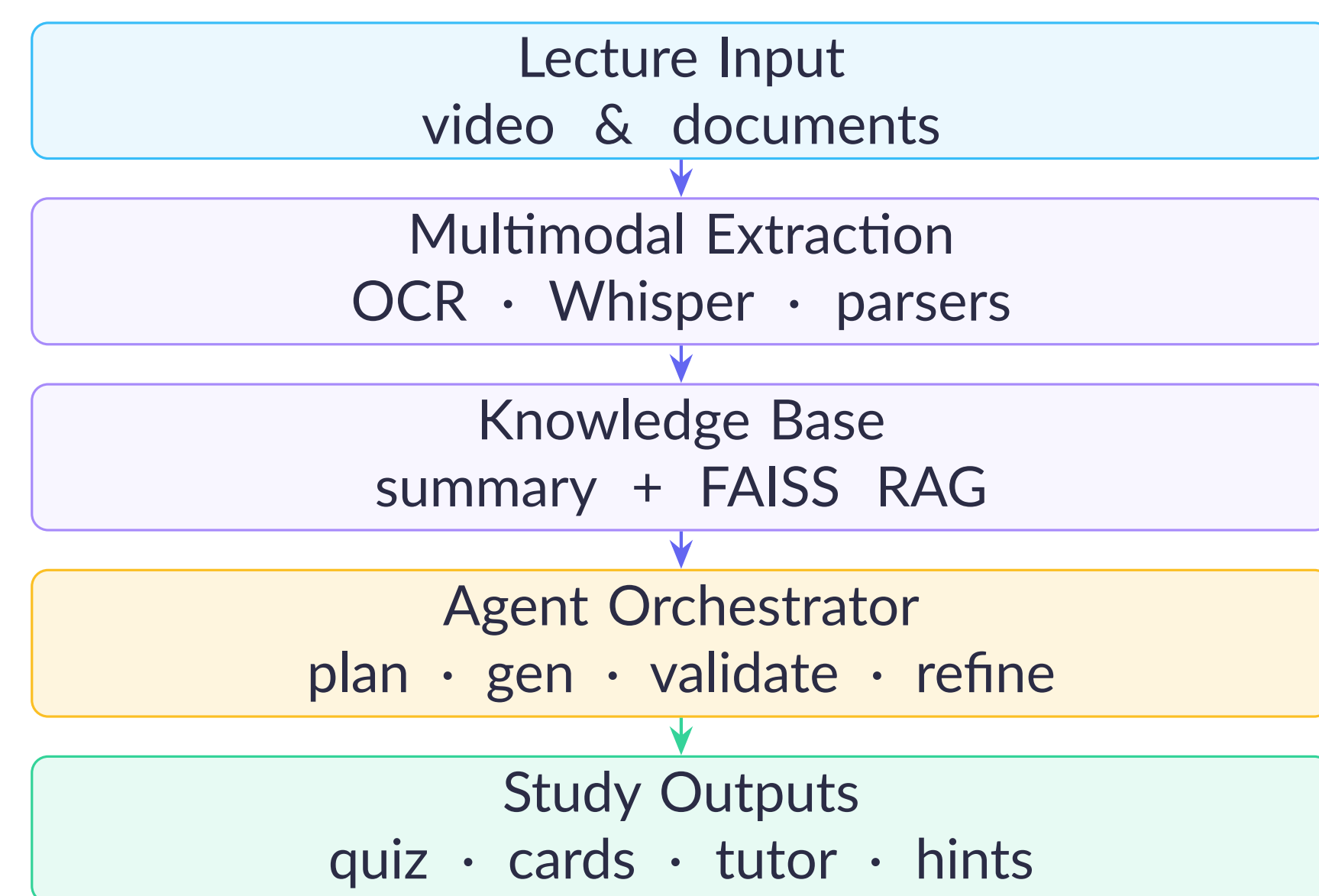


Figure 1. System pipeline: Flask on Colab → ngrok → browser UI.

Key Contributions

- Validation-gated, one-at-a-time quiz generation** that eliminates hallucinated, off-topic and duplicate questions.
- Fully local, \$0-API** multi-agent pipeline on a single free T4 GPU (Ollama + Gemma 3).
- Bilingual (EN + BN) multimodal extraction** with per-surface OCR preprocessing for slides, blackboards and whiteboards.
- Adaptive + Socratic learning** grounded with clickable lecture **timestamps** and inline video clips.

Methodology — Multimodal Extraction

Every modality is recovered and time-aligned *before* any LLM call.

- Visual / OCR (EasyOCR).** Frames sampled ~ 5 s apart, *classified* (slide / blackboard / whiteboard / scene) with per-class preprocessing (CLAHE, sharpening, Otsu, chalk inversion). **EN + BN**.
- Speech (Faster-Whisper).** GPU `float16` / CPU `int8` fallback, VAD filtering, per-segment timestamps.
- Documents.** PyMuPDF, python-docx, python-pptx (structure kept).
- Drive OCR cache** keyed by video hash — re-uploads skip re-scan.

Methodology — Multi-Agent Orchestration

A central **Orchestrator** runs specialist agents; independent agents run **concurrently**. Each follows `should_run` → `plan` → `execute` → `validate` → `retry`, and any crash is caught and logged so the pipeline **degrades gracefully**.

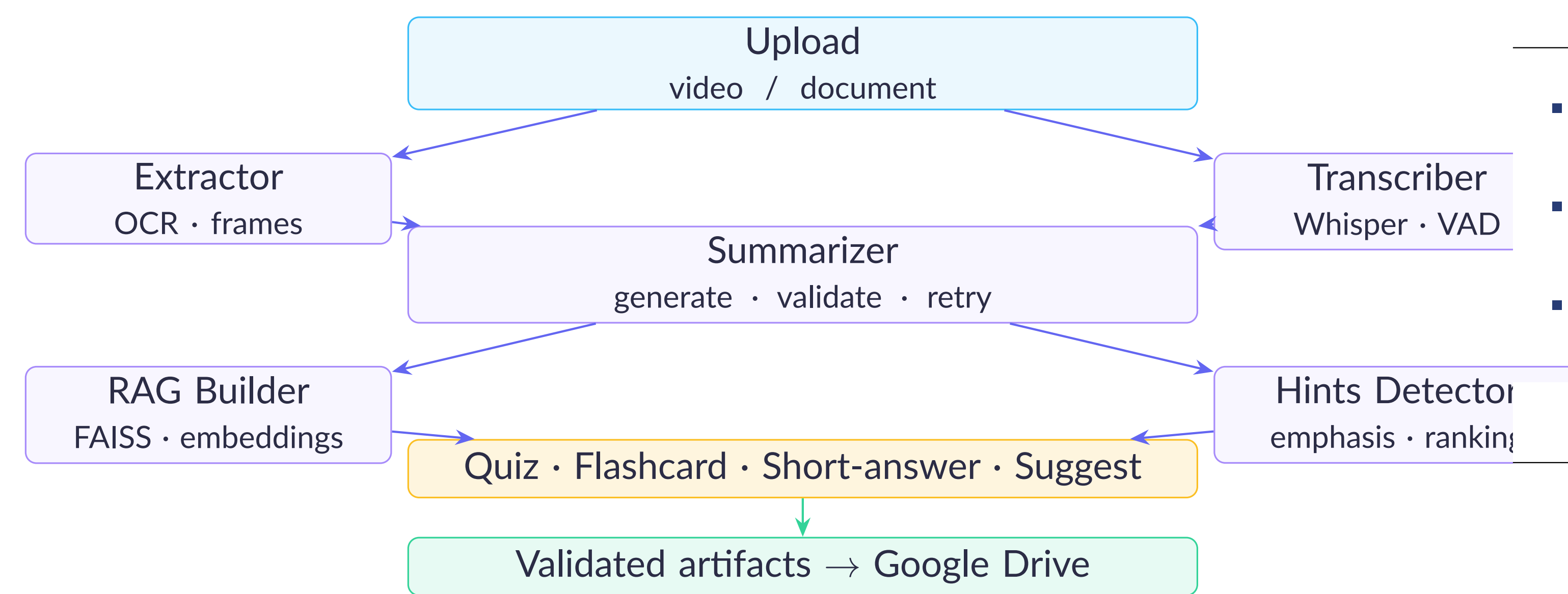
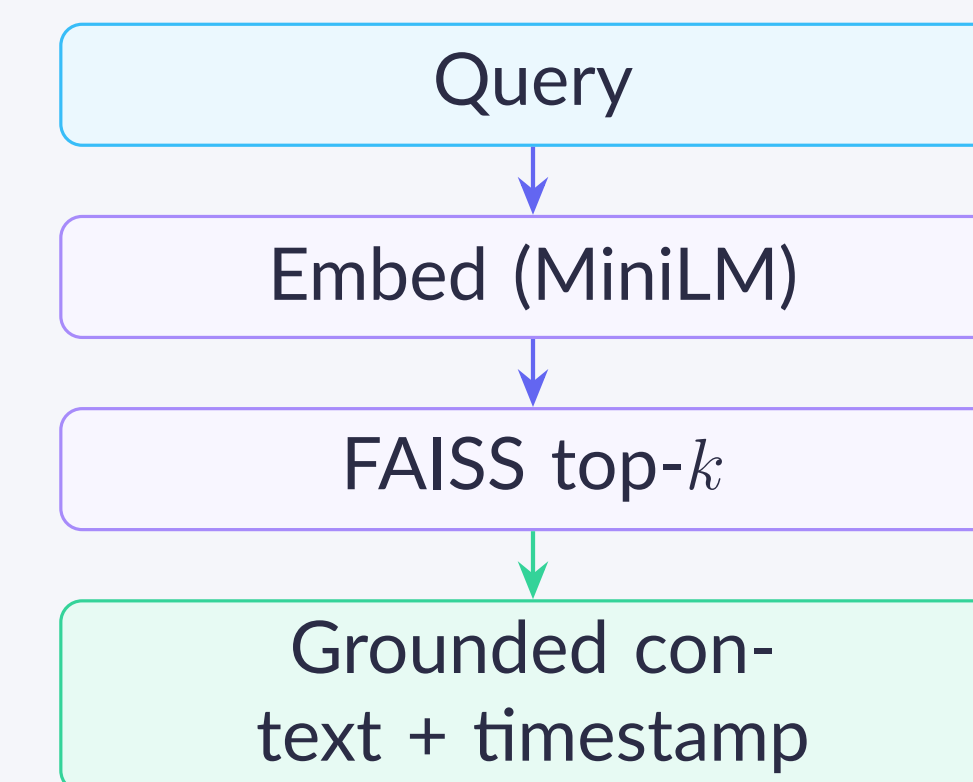


Figure 2. Orchestrated specialist agents with concurrent, fault-isolated stages.

Grounded Retrieval (RAG)

Speech and board/slide text are embedded with **Sentence-Transformers (MiniLM)** and indexed in **FAISS** (cosine). Every quiz, summary and chat answer retrieves **top- k** chunks first — anchored to real content with a clickable timestamp.



Agentic Quiz Generation

Questions are built **one at a time** and must *pass validation* before the next is generated — eliminating hallucinated, off-topic or duplicate questions. Modes: *Auto* · *Manual* (Easy / Med / Hard) · *Adaptive* (targets weak topics).

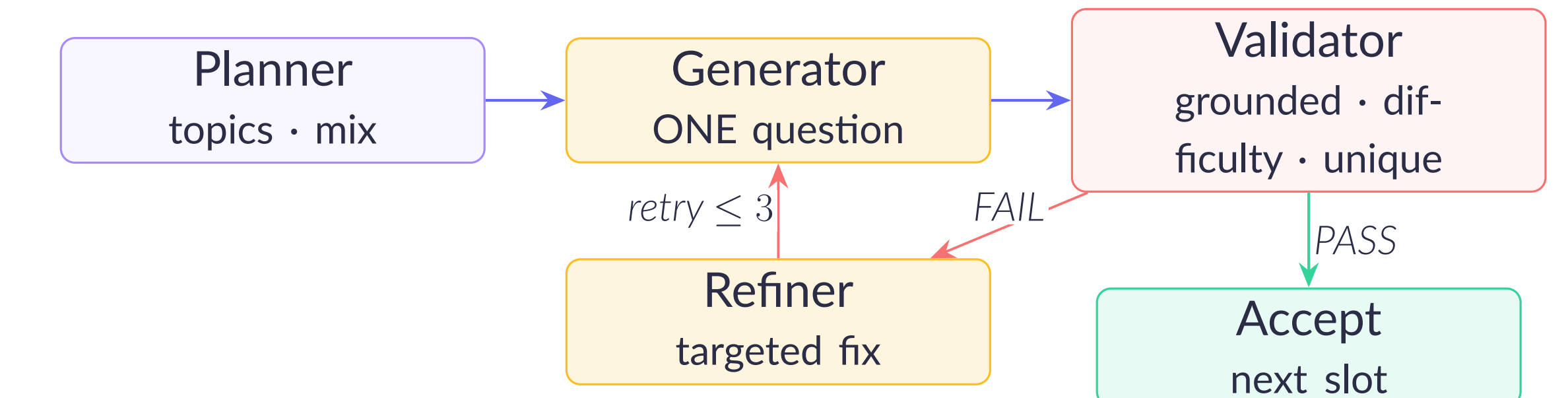


Figure 3. Generate → validate → refine loop (≤ 3 retries per slot).

Validation gate: grounded in lecture · correct difficulty · follows instructions · not a duplicate ($>85\%$ similarity rejected).

Adaptive & Socratic Learning

- Self-improving difficulty.** A Difficulty Adapter tracks per-topic accuracy, flags weak topics ($<50\%$) and steers the planner there.
- Socratic tutor.** Asks guiding questions; reveals the full answer only after 2 attempts. Cites timestamps and plays the exact **video clip** inline.
- AI short-answer grading** with partial credit and feedback.

Results & Capabilities

Capability	Outcome
Input formats	5 video + 3 doc
OCR languages	EN + BN
Per-question checks	4, ≤ 3 retries
Quiz / study types	5
Grounding	100% + timestamp
LLM hosting	local, \$0 API
Cached OCR re-runs	skipped
Agent failures	isolated

Table 1. System capabilities. Eval metrics (acceptance rate, validator-vs-human, OCR/WER, T4 runtime) to be added.

Conclusion & Future Work

LectureAssist shows that a **fully local, validation-gated multi-agent** system can turn raw lectures into grounded, adaptive study material at zero API cost. **Next:** quantitative evaluation of validator accuracy and OCR/WER, broader language support, and longitudinal learning-gain studies.

References

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